TEXORA · LECTURE SHEET

Fibres, Yarns and Fabric Structures in Technical Textiles

The raw materials and architectures from which performance is built: the fibre families used in technical applications, the yarn forms they take, and the woven, knitted, braided, nonwoven and 3D structures — with the properties each choice buys.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 2 · LEVEL 4 TERM 2

1. The design chain

Every technical textile is a chain of decisions: **fibre → yarn → structure → finishing/combination**. Properties multiply down the chain — a strong fibre in a weak structure gives a mediocre fabric, and a modest fibre in a well-designed structure can outperform it. The first three links are covered here; selection logic is Topic 3.

2. Fibres used in technical textiles

2.1 Conventional fibres in technical roles

- **Polyester (PET)** — the workhorse: good strength, low moisture uptake, UV and chemical resistance, low cost. High-tenacity grades serve tyre cord, seat belts, geotextiles, ropes.
- **Polyamide (nylon 6, 6.6)** — high toughness and abrasion resistance; airbags, parachutes, ropes, conveyor belts. Absorbs more moisture than polyester and loses some strength wet.
- **Polypropylene (PP)** — lightest common fibre (density 0.91, floats), chemically inert, cheap; geotextiles, ropes, packaging, hygiene nonwovens. Low melting point (~165 °C) limits hot uses.
- **Viscose and other cellulosics** — absorbent; hygiene and medical products, tyre cord (rayon) where dimensional behaviour suits.
- **Cotton and other natural fibres** — absorbency and comfort roles in medtech and clothtech; jute and coir in geotextiles and packaging where biodegradability is wanted.

2.2 High-performance fibres

FIBRE	HEADLINE PROPERTIES (TYPICAL)	LIMITATIONS	SIGNATURE USES

Para-aramid (e.g. Kevlar-type)	Very high tenacity (~2.8–3.0 GPa class), heat resistant, no melting	Degrades in UV; absorbs moisture; poor compression strength	Ballistics, cut/heat protection, ropes, composites
Meta-aramid (e.g. Nomex-type)	Flame and heat resistance, lower strength than para-aramid	Moderate tenacity	Firefighter clothing, insulation, honeycomb
HMPE / UHMWPE	Highest strength-to-weight of commercial fibres; floats; excellent abrasion and cut resistance	Melts ~145–150 °C; creeps under sustained load	Mooring ropes, ballistic panels, cut gloves
Carbon	Very high modulus and strength, light, electrically conductive, fatigue resistant	Brittle, expensive, black only	Aerospace/automotive composites, pressure vessels
Glass (E-glass)	High strength, incombustible, cheap for its stiffness	Heavy (density 2.55), brittle, abrades	Composites, insulation, filtration, buildtech
Ceramic / basalt	Very high temperature capability	Brittle, costly	High-temperature insulation and filtration
PTFE	Chemically almost inert, wide temperature range, low friction	Low strength, expensive, difficult to process	Chemical filtration, architectural membranes, sewing thread
PPS, PEEK, PI	Heat and chemical resistance for hot filtration and engineering uses	Expensive	Hot-gas filter bags, aerospace parts
Stainless steel / metal fibres	Conductivity, heat resistance, cut resistance	Heavy, costly	Antistatic and conductive fabrics, heating elements

Table 1 — High-performance fibre families. Values are typical textbook ranges for orientation, not specifications; any real design uses the maker's data sheet.

3. Yarn forms

- **Monofilament** — a single thick filament: meshes, screens, surgical mesh, fishing lines. Round, smooth, exact diameter.

- **Multifilament yarn** — many fine filaments twisted or textured together: the main load-bearing yarn of technical fabrics; high-tenacity versions for belts, cords, airbags.
- **Spun yarn** — from staple fibres; used where surface hairiness, friction or absorbency is wanted (some filter and polishing products).
- **Tape / slit-film yarn** — slit polypropylene or polyethylene film: woven sacks, carpet backing, artificial turf, geotextile tapes.
- **Textured and bulked yarns** — crimped filaments giving bulk and cover: upholstery, insulation, filtration depth media.
- **Hybrid and commingled yarns** — two materials in one yarn (e.g. glass + thermoplastic for composites, core-spun yarns with an elastic or high-strength core): the yarn itself becomes a designed structure.
- **Cords, ropes and braided preforms** — yarns assembled into load paths: tyre cord, mooring lines, composite preforms.

4. Fabric structures

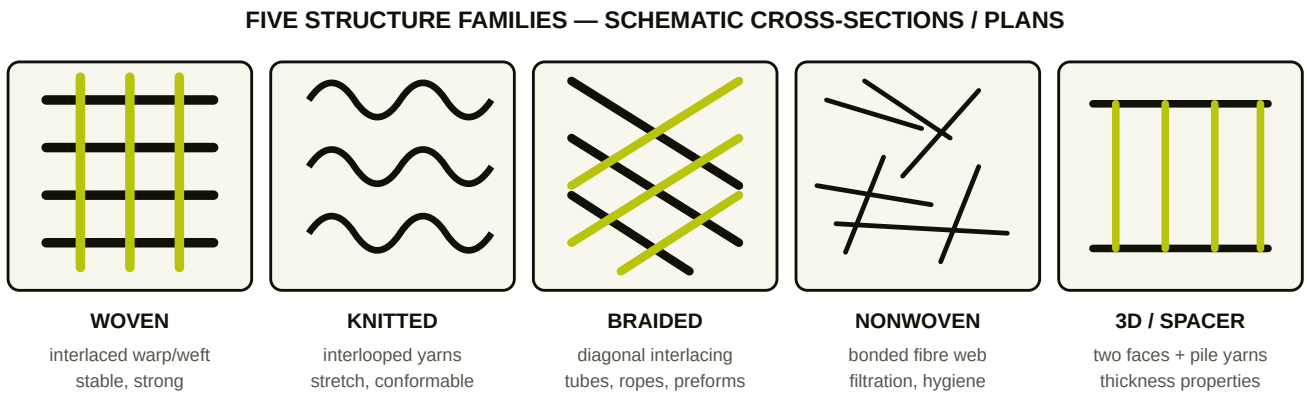


Figure 1 — The five structure families used in technical textiles, drawn schematically (not to scale).

STRUCTURE	CHARACTER	TYPICAL TECHNICAL USES
Woven	Dimensionally stable, strong in two directions, controllable pore size	Geotextiles, filter fabrics, airbags, belts, reinforcement fabrics
Knitted (weft and warp)	Stretch and conformability; warp knits are stable and run-resistant	Spacer fabrics, medical meshes, sporttech, shade nets, composite preforms
Braided	Tubular or flat, excellent torsion behaviour, expands/contracts radially	Ropes, sutures, catheters, composite preforms over mandrels

Nonwoven	Fibre web bonded by needling, heat, chemicals or water jets; cheapest route to area properties	Filtration, hygiene, geotextiles, insulation, wipes, interlinings
3D structures	Thickness-direction yarns (spacer, 3D woven, multiaxial)	Composites reinforcement, cushioning, protective padding, scaffolds
Coated / laminated / composite	A structure combined with a film, coating or matrix for barrier or stiffness	Membranes, tarpaulins, airbags, textile composites (Topic 5)

Table 2 — Structure families and the properties they characteristically deliver.

5. Worked example — from linear density to fibre diameter and surface area

Fine fibres filter and bond better largely because of their surface area. For a cylindrical fibre of density ρ and linear density *t* (mass per unit length), the cross-sectional area is A = t/ρ and the diameter follows from A = πd²/4. The specific surface area (surface per unit mass) of a cylinder is:

$$SSA = 4 / (\rho \cdot d)$$

WORKED EXAMPLE — POLYESTER FILAMENT, 1.4 DTEX

t = 0.14 tex = 0.14×10^{-6} kg/m; ρ(PET) = 1,380 kg/m³.

A = t/ρ = 0.14×10^{-6} / 1,380 = 1.01×10^{-10} m².

d = √(4A/π) = √($4 \times 1.01 \times 10^{-10}$ / π) ≈ **11.4 µm**.

SSA = 4 / (1,380 × 11.4×10^{-6}) ≈ 255 m²/kg ≈ **0.26 m²/g**.

Interpretation: even an ordinary apparel filament presents about a quarter of a square metre of surface per gram — which is why surface finishes work at gram-per-kilogram dosages, and why going finer (Topic 8, nanofibres) multiplies surface effects so strongly.

Glossary

Tenacity — breaking force per unit linear density (N/tex or cN/tex); the honest strength measure for comparing fibres of different fineness. **Modulus** — stiffness, stress per unit strain in the initial region. **High-tenacity yarn** — filament yarn produced for elevated tenacity by high draw ratio and controlled polymer quality. **Preform** — a textile structure shaped ready for composite moulding. **Multiaxial fabric** — layers of straight yarns in several directions, stitched together, used as composite reinforcement.

Key points

• Performance is built in a chain: fibre → yarn → structure → finish; weakness at any link caps the product.
• Conventional fibres (PET, PA, PP) do most technical work; high-performance fibres (aramid, HMPE, carbon, glass, PTFE) serve heat, strength-weight and chemical extremes — at a price.
• Yarn form is a design variable: monofilament, multifilament, tape, hybrid and core-spun yarns are different tools.
• Structure families trade stability (woven) against conformability (knitted), tubular form (braid), cheap area properties (nonwoven) and thickness properties (3D).
• Fibre fineness controls surface area: SSA = 4/(ρd) — the quantitative root of filtration and finishing behaviour.

Practice

1. Repeat the worked example for a polypropylene filament of 2.2 dtex (ρ = 910 kg/m³): find its diameter and specific surface area.
2. A filter bag must survive hot, chemically aggressive flue gas. Using Table 1, shortlist two candidate fibres and state the deciding property of each.
3. Explain why a seat belt is woven from high-tenacity multifilament polyester rather than from spun polyester yarn of the same total linear density.

Answer key

1. $t = 0.22 \times 10^{-6}$ kg/m. $A = 0.22 \times 10^{-6}/910 = 2.42 \times 10^{-10}$ m². $d = \sqrt{(4A/\pi)} \approx$ **17.5 µm**. SSA = $4/(910 \times 17.5 \times 10^{-6}) \approx 251$ m²/kg ≈ **0.25 m²/g** — PP's lower density offsets its coarser filament, giving nearly the same surface per gram as the PET example.
2. **PPS** — heat and chemical resistance for hot-gas filtration; **PTFE** — near-total chemical inertness across a wide temperature range. (Glass is a defensible third for temperature, with brittleness as its caution.) The deciding properties are temperature capability and chemical resistance, not tensile strength.
3. Multifilament yarn packs continuous, highly drawn filaments: tenacity is higher and, critically, *uniform* — no weak places from fibre-length statistics — and abrasion is shared across many filaments. A spun yarn's strength is limited by fibre-to-fibre friction and its weakest cross-section; in a safety product whose failure injures people, the continuous-filament route gives the predictable breaking force the specification demands.